

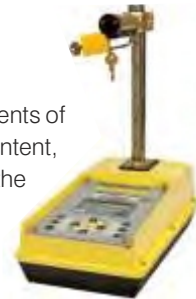
Soil Field & Laboratory

Compaction, Moisture Testing

Nuclear Density Gauge (NDG)

Moisture/Density Gauges use an advanced micro-processor-based technology to provide highly-accurate measurements of density and moisture that are automatically computed for direct readouts of wet density, dry density, moisture content, percent of moisture, percent of compaction (Proctor or Marshall), void ratio and air voids. The NDG can be used to test the degree of compaction of practically any paving material such as soil, aggregate, asphalt, concrete, even flyash.

Manufacturer approved re-calibration facilities are available to ensure that gauge accuracy is maintained over several years of use.



Electrical Density Gauge (EDG)

The Electrical Density Gauge (EDG) is a nuclear-free alternative for determining the moisture and density of compacted soils used in road beds and foundations. The EDG is a portable, battery-powered instrument capable of being used anywhere without the concerns and regulations associated with nuclear safety. Its user-friendly, step-by-step menu guides the user through each step of the testing procedure and cautions the user when values do not correspond to established curves for the material being tested.

Easy-to-use, the EDG can be used as a construction aid to monitor day-to-day compaction operations by providing performance and measurement results highly comparable to those achieved with traditional methods, including the nuclear gauge and/or a sand-cone and oven moisture test combination.

Strength Measurement

Electronic Plate Bearing Tester-Static

The Electronic Plate Load Device allows simple determination of the load-bearing and deformation capacity of soils. The load settlement lines and the modulus of deformation E_{v1} , E_{v2} are determined to DIN 18134:2001-09 or E DIN 18134:2010-04 (plate load test). Thanks to the splash proof housing, external buttons and a backlit display, it can also be used under inclement ambient conditions. Test logs can be printed out directly on site. The results saved on an SD card during the test can be transferred to the PC and are available for further processing under Microsoft Excel®



Dynamic Light Weight Deflectometer/ Drop Weight Tester

This dynamic plate load test is used in earthworks, civil engineering, road construction and rail track construction to determine the load-bearing capacity and to assess the compaction of soils, loose base courses and soil improvement in accordance with TP BF – StB Part B 8.3.

The test method is suitable for coarse-grained and mixed-grained soils up to a maximum grain size of 63 mm. The test result is displayed in MN/m^2 . The measuring range of the light drop weight tester lies between 15 and 70 MN/m^2 . With the 1.5x impact load achieved with the 15 kg loading device, the measuring range is extended to 70 – 105 MN/m^2 . This is required for heavy base courses in road beds and for ballast.

Geogauge

The GeoGauge is a unique, QC/QA field tool that can be used to measure the uniformity of unbound pavement layers by measuring the variability in stiffness throughout a structure. It is an excellent tool for identifying construction anomalies that would otherwise go undetected during construction where only density or percent compaction measurements were used. GeoGauge can be used to verify that the stiffness/modulus values assumed in the design specifications of new or rehabilitated pavement structures are met.

This dynamic technology used by the GeoGauge simulates real in-use conditions. This factor allows the GeoGauge to directly measure in-place engineering properties during the construction process.



Soil Strength Classifier

Hand-lever operated, soil strength classifier provides fast manual approximations of unconfined compressive strength on many soil types in the field or lab. Platen accepts up to 2" (51mm) sample diameters and up to 4-1/2" (114mm) lengths. Reads loads on 4-1/4" (108mm) dial face with $\pm 1\%$ of dial capacity as held with second maximum-reading pointer.

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In-situ CBR Test

The 50KN mechanical jack is designed to perform C.B.R. tests in-situ (BS 1377, ASTM D 4429) but may also be used for plate bearing tests. The aluminium housing encloses a worm screw reducer. A quick-release key allows the rapid adjustment of the plunger.



Motorized Dynamic Penetrometer - 30 Kg

The 30 Kg motorized dynamic penetrometer is supplied with a 30 kg hammer having a 20 cm fall and cones with 10 sq. cm² section. The hydraulic drive unit slides along a guide bracket mounted on a robust foot plate. The support frame has an overall weight of around 30kg and may be disassembled so as to reduce size for transport.



Dual Mass Dynamic Cone Penetrometers - Quick Connect Ends

Developed by the Army Corps of Engineers, Dynamic Cone Penetrometers (DCPs) provide a low-cost, efficient test method for quickly determining in-situ CBR values of pavement base, sub base and sub grades. DCPs are primarily used to determine in-place soil shear strength in road construction with CBR values from less than 0.5 to 100% and bearing values ranging from 430 to 10,800 psf. They can readily be used for depths from 30 inches to 6 foot with optional drive rods and extensions. The DCPs comply with ASTM D6951 specifications and come with a chart to compute CBR values, as well as an Excel spreadsheet template, which automatically charts the test results.

Other variants of Dynamic Cone Penetrometers are also available.

Static Penetrometer 100KN - Trailer Mounted

The machine is mounted on a single axle trailer complete with electric installation, lights, overrun brake (which automatically comes into operation on braking the towing vehicle) and hand brake. The group may be attested for road circulation. Total weight is approx. 1500 kg. During transportation the hydraulic drive ram is lowered between the two vertical support columns and inclined in a horizontal position so as to reduce the total height. Anchorage is provided by the classic system of spiral anchors driven into the ground using a hydraulic wrench which requires the presence of two operators.



Static Cone Penetrometer, Digital

The digital, static cone penetrometer (DSCP), with its large, digital readout, makes testing and recording readings easy. The DSCP can be used to evaluate soil consistency by determining the soil's level of compaction and/or the bearing capacity. The DSCP is especially useful in evaluating shallow foundations and pavement subgrades where fine-grained and soft soils are being used. The DSCP can also be used to aid technicians in quickly selecting sites for testing, as well as correlating with other tests, based on local conditions, such as standard or modified Proctor, CBR or bearing capacity. The DSCP uses a dual-rod design, which eliminates the need to correct for soil friction on the rod as the cone is pushed through the material. Penetration resistance is read directly from the cone tip and registered on the digital display.

Other variants of Static Cone Penetrometers are also available.



Light-weight Static/ Dynamic Self Anchoring Penetrometer

This instrument is able to make both static and dynamic penetrometer tests. To change from one test to another, it is only necessary to change the cone. In the static form, no preliminary operation to anchor the machine to the ground is necessary because the special cone is self-anchoring; between one static reading and the next, the sounding rod advancement is achieved by percussion.

In the dynamic configuration, normal fixed or "lost" cones (with diameter larger than the sounding rods) are used. If the hole wall is unstable, the parasitic friction that generates along the rod string may be eliminated by the injection of liquids. The functioning of the unit is fully hydraulic both during the driving and extraction phase.



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Soil Penetrometer, Pocket/ Dial-Type

A sophisticated pocket penetrometer offering greater capacity and sensitivity than standard available pocket type penetrometers. Maximum value is retained on the dial until released via push-button. Inner dial scale 0 to 6.0, with 0.1 divisions in tsf and kg/cm². Outer scale gives load strength over 0 to 11.0 with 0.1 divisions in kg. This reading is used with charts (included) to estimate safe bearing pressures, depending on plunger used and soil type. Values indicated relate to the standard 0.25" dia. plunger.

Other variants of Pocket Penetrometers are also available



Geovane Tester

The Geovane is a hand-held instrument used for determining soil shear strength providing the reading in kPa. A 19mm vane blade is screwed into the base of the Geovane and the vane is pushed into the soil and rotated and reading is read off the face when the soil fails. The pointer stays in place when failure occurs allowing you to look up the indicated reading on the supplied calibration chart to get your reading in kPa from zero to 200.

Torvane Shear Tester

This device is recommended for the rapid determination of shear strength of cohesive soils, in either the field or laboratory. It permits the determination of a large number of strength values with different orientation of failure planes. It is simple to use with sample trimming eliminated. Extensive laboratory testing indicates excellent agreement between the unconfined compression test and the shear tester.



Dielectric/ Electrical Conductivity

Percometer

The Percometer is a reliable, accurate, lightweight and easy-to-use instrument for measuring the dielectric value (er), electrical conductivity (J) and temperature of materials indoors and outdoors. Dielectric value is an indication of the volumetric moisture content and the state of molecular bonding in a material. Electrical conductivity is a reflection of ionic concentration, water content and temperature. The ability of the Percometer to measure these values has made it an extremely versatile tool in a multitude of tasks.

The Percometer has been very successful in providing data for The Tube Suction Test (TST), which was developed by the Finnish National Road Administration and the Texas Transportation Institute (TTI) for assessing the moisture susceptibility of granular base materials.



Field Resistivity

Resistivity Meter

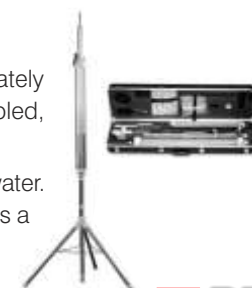
The resistivity meter can be easily used to measure resistivity of soil in-situ using the 2, 3, or 4 electrode method or with the soil box to measure resistivity of soil or water in the range of distilled to seawater. It has a wide range (0.01 ohm to 1.1 megohm) in 8 ranges with 10% overlap. It is fully temperature stabilized over temperature ranges encountered in all field conditions; is insensitive to A.C. or D.C. ground currents; can be used safely in wet ground and comes with (1) 12-volt battery and shoulder carrying strap.

Hydraulic Conductivity

Guelph Permeameter Kit

The Guelph permeameter, supporting the ASTM D5126, is an easy-to-use instrument for quickly and accurately measuring in-situ, hydraulic conductivity in all types of soils. The equipment can be transported, assembled, and operated easily by one person.

Measurements can be made in 1/2 to 2 hours, depending on soil type, and require only about 2.5 liters of water. Measurements can be made in the range of 15 to 75cm below the soil surface. The Guelph permeameter is a complete kit consisting of the permeameter, field tripod, well auger, well preparation and cleanup tools, collapsible water container, and vacuum test hand pump, all packaged in a durable carrying case.





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Sampling

Soil Sampling Kit

Basic sampling kit provides everything you'll need to auger to a target depth as deep as 12ft and obtain a relatively undisturbed soil core sample for soil testing. The kits are used worldwide by construction companies, consultants and engineering firms for site investigations.

Other sampling kit options are also available. Please contact us for more details.



Shelby Tubes

Shelby tube samplers are thin-walled, hollow steel tubes, which are driven into the ground to extract a relatively undisturbed soil sample for use in laboratory tests used to determine density, permeability, compressibility and strength.



Soil Sampling Auger Tube Set

Auger and sampling tube with (2) 12" (305mm) extensions. Set contains auger and sampling tube. Components are plated steel to resist abrasive action of soil. Includes 1.25" (25mm) dia. by 12.5" (318mm) long auger, 1" (approx. 25mm) OD by 12.5" (318mm) long sampling tube, two 12" (305mm) extension rods, handle and a fiberboard carrying case.



Power Mechanical Earth Drill

For soil sampling, construction and more, earth augers are gasoline powered and portable. Using one operator, they allow fast and easy drilling. They feature snap-on or screw-on augers and extensions. Engine mounted on wheel-base carrier keeps motor noise and fumes away from the operator. Torque tube eliminates counter torque. Features such as the carrier and torque tube offer greater mobility, beneficial in areas inaccessible to large equipment, on slopes and horizontal drilling, various standard sampling methods to obtain material samples.

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California Bearing Ratio

Humboldt MasterLoader

The Humboldt MasterLoader CBR/LBR Solution supports ASTM D1883, AASHTO T193, and BS 1337 Part 4. The Humboldt MasterLoader is the most versatile load frame available today. It's the perfect solution for labs requiring versatility to perform a variety of tests in addition to CBR/LBR. The MasterLoader provides an internal 4-channel data logger, which allows it to be used as a standalone unit or as part of a computer-controlled system, the MasterLoader provides the user with fully-automatic test performance allowing unattended operation while controlling motor start/stop, speed selection and test data acquisition.

Load Cell and LSCT displacement sensors and reporting software are available for data download and reporting.



Humboldt CBR/LBR Specific Automatic Load Frame

A fully automatic, single-speed load frame, (1.27mm/minute), designed for those who want a high-quality, application-specific load frame that provides simple operation with built-in data acquisition capabilities. Provides two channels with integral data acquisition to accommodate a load cell and transducer for CBR/LBR testing.

Load Cell and LSCT displacement sensors and reporting software are available for data download and reporting.

HMTS Reporting Software, CBR/LBR Module

The CBR/LBR Module provides a simple, test-specific interface to control CBR/LBR test operations and automatically record data while also displaying it in real-time tables and graphs. Technicians can be freed-up for other duties with the assurance that all test data is being collected and saved.

Permeability/ Hydraulic Conductivity



Constant Head Permeameter Cells

The Constant Head Permeameter Cells supports ASTM D2434 and BS 1377 Part 5, used to determine the coefficient of permeability by the constant head method for laminar flow of water through granular soils. Two manometer ports are grooved & screened on the inside. Distance between ports is always equal to diameter. 100 mesh screens used to prevent migration of material through valves & tubing during test. Acrylic chamber permits viewing sample. Spring incorporated into top cap to apply 5-10 lbs. force against screen to prevent soil density changes.

Falling Head Permeameter Cells

The Falling Head Permeameter Cells supports ASTM D2434 and BS 1377 Part 5, for use with granular soils in determining the coefficient of permeability via the falling-head method for laminar flow of water. Compact and portable unit includes a spring incorporated into top cap to apply 5-10 lbs. force against screen to prevent soil density changes.

Shelby Tube Permeameter

Allows you to perform permeability tests within a Shelby tube without removing the sample. Ideal for cohesionless materials and sands. It is suggested that the specimen be ejected at test conclusion and examined for voids or large aggregate, which possibly could affect the test results.



Compaction Permeameters

The Compaction Permeameters supports ASTM D5856. For determining permeability of clay, sand, and gravel soils. Uses Proctor plasticity compaction method. Two-piece cylindrical mold includes: upper plate with valve; water inlet/outlet; filter base plate; and filter stones.



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Flexible Wall Permeability Apparatus

Flexible Wall Permeability Cells

The flexible wall permeability cells meet ASTM D5084 and are constructed of high-quality materials throughout for long-lasting performance. The cell top and base are precision machined from 6061 T6 Aluminium and then hard-coated and Teflon-impregnated for a durable finish. To facilitate sample setup, the chamber and cell top can be quickly and easily removed by loosening the three knobs that hold the upper assembly to the base.

The cell has five, no-volume-change, valves aligned along the front of the cell for maximum convenience. Two valves handle top drainage, two valves handle bottom drainage, and one valve handles filling/emptying and providing confining pressure to the cell. Brass valves are standard with these cells, but stainless steel valves (stainless steel is typically used with hazardous materials) are also available as option.



Humboldt FlexPanels

Humboldt FlexPanels provide a simple and highly efficient distribution system for providing air, water and de-aired water for use in permeability and triaxial testing applications. The FlexPanel's simple, straight-forward configuration, with its integral burettes provides a condensed/compact design that takes up less counter space than competing systems with air/water bladders.

The long, narrow burette design of FlexPanels provide faster test processing times when compared to larger, shorter burette systems, while providing the same volume. This is due to the reduced amount of meniscus formation in the narrower burettes, which allows the water level to drop faster, resulting in faster readings. In addition, the use of longer/narrower burettes and a scale graduation of 0.02ml, also provides an easier-to-read and more accurate scale.

FlexPanels also feature a bias regulator and bridge. The bias regulator maintains the differential pressure when confining and back pressures are increased. The bridge delivers simultaneous control of base and top pressures through the use of just one regulator. This feature minimizes operator time and reduces the margin of error in opening and adjusting regulators during a test



Toxic Interface Unit

Safe and convenient means of performing permeability tests of corrosive or toxic permeants. Flexible fluoroelastomer bladder accumulator interfaces between control panel and sample drains on permeameter. Serves as a fluid separator to prevent permeant from entering control panel. Also prevents contact of air with permeant, thus no toxic or corrosive vapors can escape into lab

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Consolidation

Humboldt ConMatic IPC, Automated Consolidation System

Conforms to ASTM: D2435, D4546, AASHTO: T216, BS:1377:5. The ConMatic IPC is a fully-automated, incremental pressure controller for performing incremental consolidation and one-dimensional swell tests. The ConMatic IPC allows consolidation and constant load and volume swell tests to be run automatically, freeing up technicians for other tasks and reducing the duration of the testing procedures by more than half—effectively saving time and manpower and increasing lab profitability. One ConMatic automated system can replace the production of several manual machines—running incremental consolidation tests according to ASTM D2435 Method B, where successive load increments are applied after 100% primary consolidation. Once a sample has been placed onto the test platform and the test conditions set, the ConMatic IPC, used in conjunction with a computer and Humboldt's HMTS software, performs all consolidation



GCTS Consolidation Apparatus

The GCTS Consolidation Apparatus is capable of performing various tests such as: controlled gradient, constant rate of strain, incremental consolidation, permeability, and conventional consolidation tests with extreme accuracy and ease. Features:

- Constant Rate of Strain (CSR), Constant Gradient (CG), and conventional Consolidation Tests
- Electro-Pneumatic Servo Control
- 10 kN load capacity
- Automatic data logging and report generation
- Custom Specimen sizes available

Direct/ Residual and Residual Ring Shear Tests

Pneumatic Direct/Residual Shear Apparatus

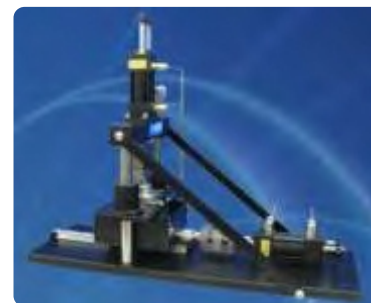
Conforms to ASTM: D3080; AASHTO: T236, BS:1377:7. The Humboldt Pneumatic Direct/Residual Shear Apparatus utilizes the pneumatic loading concept for applying the vertical load to the sample. In doing so, this self-contained model eliminates the need for cumbersome loading weights used in dead weight-type systems. The microprocessor-based system features a stepper motor drive system, large display, touch-sensitive keypad and forward/reverse travel limit switches. Through the use of a built-in 4-channel data acquisition system, the operator can preset the logging condition for the test.



Electro-Hydraulic Direct Shear System

This system features electro-hydraulic closed-loop digital servo control of the shear and normal loads for test automation. Loads or deformations for both the shear and normal actuators can be prescribed for automatically performing conventional direct shear tests as well as more advanced tests. This system can be easily programmed to perform tests such as the constant normal stiffness test where the normal load is a function of a prescribed stiffness for simulating actual compressibility of a ground shear plane (e.g. soil-pile interaction).

A major advantage over conventional motorized system is that this system eliminates vibrations that can disturb or densify granular specimens.



Residual Ring Shear Test

The GCTS Residual Ring Shear System is ideal for determining the drained residual strength of unsaturated soil. It is a fully automatic, electro-pneumatic, servo-controlled system that provides accurate results while remaining easy to use. This system includes our Harmonic Drive™ motor technology which allows for continuous rotation without backlash.

The Residual Ring System can also perform multi-stage tests, allowing a single specimen to be used to determine the residual strength under multiple normal loads. It can also be used to test saturated soil specimens.

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Triaxial Testing

Static Triaxial Systems

Humboldt provides an extensive line of triaxial testing equipment solutions for today's soil labs. At the heart of our triaxial testing equipment is the Humboldt Concept of providing a modular system of interchangeable, stand-alone components that when combined create highly-versatile systems. This modular concept allows you to easily create a custom solution for your needs, as well as having the ability of taking advantage of upgrades and new technology, while not being locked into an obsolete proprietary system.

The Humboldt triaxial systems are based around automated electro-mechanical load frames, HMTS software with triaxial-specific software modules and three different pressure control solutions.



Automated Pressure Control Triaxial System

Designed for those who want the ultimate in control of their triaxial testing, Humboldt's automated pressure control triaxial system is a computer-controlled system specifically designed for soil testing laboratories conducting UU, CU and CD Triaxial tests, as well as unconfined compression. It is perfect for large, high-volume labs, as well as those who want to utilize technology to increase staff efficiencies and testing accuracy. This system provides complete control of the testing process including data acquisition.

Manual pressure controlled systems are also available from Humboldt. Please contact us for more information.



Servo Controlled Dynamic/ Cyclic Triaxial Systems



Servo Pneumatic Soil Triaxial Testing System

The GCTS Pneumatic Soil Triaxial Testing System is intended for performing conventional triaxial tests as well as more advanced procedures, such as the K_0 test and stress/strain path tests. This system also provides the necessary versatility to easily perform dynamic tests for liquefaction, resilient modulus, and cyclic strength. The triaxial testing system includes all of the necessary software modules to perform the above tests.

The system includes an automatic ball valve to automatically perform B-value checks as well as automatically open or close drainage paths as required by different test stages. It incorporates high resolution control of the pneumatic loader, allowing it to obtain clear stress-strain curves and apply load at very low rates.

Summary results for individual test specimens can be generated in report format and files can be merged to produce a Mohr envelope plot for up to five specimens. The software will also automatically calculate the shear modulus and hysteretic damping for dynamic tests. Data acquired from all the software modules can be easily transferred to other programs like spreadsheets, databases, etc.

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Servo Hydraulic Soil Triaxial System

The GCTS hydraulic triaxial testing system is intended for performing dynamic tests for liquefaction, resilient modulus, cyclic strength, complex modulus, and other dynamic triaxial tests including synchronized cyclic axial and confining stress loading. This system also provides the necessary versatility to automatically perform conventional triaxial tests as well as advanced procedures such as stress or strain path.

The standard system includes all the necessary hardware and software to automatically perform all triaxial stages including saturation, consolidation, and easily created cyclic and static customized test procedures. It allows creation of an unlimited variety of waveforms including user-generated profiles such as a digitized earthquake record. The system software provides advice for the user as to how to perform testing tasks.



Large Scale Servo Hydraulic Triaxial System

The GCTS Large-Scale Cyclic Soil Triaxial System is intended for performing dynamic tests on soils with large particle sizes, such as gravely soils and railway ballasts. The triaxial cell accepts 300 mm diameter samples with heights up to 700 mm. This system is typically used to perform tests for liquefaction, resilient modulus, cyclic strength, complex modulus, synchronized cyclic axial, confining stress loading and other dynamic triaxial tests.

Extra Large-Scale Soil Triaxial System

The GCTS Extra Large-Scale Soil Triaxial System is intended for performing tests on soils with large particle sizes, such as rock backfills and gravely soils. The triaxial cell accepts 1,000 mm diameter samples with heights up to 2,000 mm. This system also provides the necessary versatility to automatically perform conventional triaxial tests as well as more advanced procedures, such as stress or strain path. In addition, the STX-2000 has the capability of performing dynamic triaxial tests.

The system includes all hardware needed to create and load a large specimen into the testing chamber without need of a forklift. It also includes a dynamic compaction frame, which can create the specimen using a 200 kN axial load and at frequencies of up to 100 Hz.



True Triaxial/ Polyaxial Testing

The GCTS Poly-Axial Testing System includes four hydraulic load rams, each equipped with LVDT's for independent computer control of each platen. Stress or strain can be used as the feedback control. With this configuration, the specimen can remain centered, minimizing end platen friction. This is accomplished by controlling the deformation rate of one LVDT on each plane and forcing the opposing LVDT on each plane to mimic the deformation rate of the controlled LVDT.



Simple Shear Testing

The design of the GCTS Simple Shear Testing System (SSH-100) is based on our external cell wall triaxial system. The apparatus has a fixed top and a sliding bottom base mounted on special linear bearings to help with specimen loading and unloading. The system also has a stiff internal support to minimize lateral compliance of the top cap.

One of the main advantages of the GCTS Simple Shear Testing System is that the apparatus does not require a reinforced membrane, as lateral support is provided via the confining pressure, which can be servo-controlled. This means that consolidation can be done at K values other than K0. However, this device also includes Teflon-coated thin rings to externally reinforce the standard latex membranes to ensure K0 consolidation.

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Resilient Modulus Testing

The GCTS Resilient Modulus Testing System boasts a modular design which allows the system to be configured to test soil in a variety of modes. This system is based on the GCTS SCON Digital Signal Conditioning and Controller and can be supplied with a pneumatic or hydraulic loader.

The GCTS Resilient Modulus software features built in AASTHO, SHRP, and NCHRP test sequences and the capability to specify user-defined sequences. Contact stress is automatically adjusted according to each procedure as selected.

The deformation ratio of the two sensors, R_v (to ensure that the two deformation sensors are in agreement), and resilient modulus, M_r , are also calculated in real-time. During export, curve fitting is done to match the results to models that predict M_r as a function of σ'_m , σ'_d , and cell pressure (CP). Four different functions are calculated automatically, as required by AASTHO and NCHRP procedures.



The system can test specimens with a 70 mm, 100 mm, or 150 mm diameter, although it is recommended to use a smaller system for 70-mm diameter specimens, such as the GCTS Electro-Pneumatic Resilient Modulus System. All standard systems include the required dual external deformation sensors with supports and internal/external load cell.

Resonant Column/ Torsional Shear Tests



Resonant Column/Torsional Shear Testing System

The Resonant Column/Torsional Shear Testing System combines the features of both resonant column and torsional shear devices into one elegant system, enabling it to evaluate the effects of a number of soil parameters, such as void ratio, confining pressure, strain amplitude, number of load cycles on shear modulus and material damping. By utilizing all available options, this system is capable of the full spectrum of modulus measurement, ranging from ultra-low strains to high strains with significant overlap.

For the resonant column test, a torsional drive is used to vibrate the top of the soil specimen at first-mode resonance at frequencies of up to 500 Hz while the bottom is fixed. A non-contacting rotational transducer is attached to the top platen monitor the torsional motion directly. The non-contacting sensor is located on extended arms to amplify the rotational deformation and provides a shear strain resolution of 10^{-6} . An optional accelerometer can also be installed to measure large shear strains.

The system can also perform closed-loop torsional shear tests (stress or strain controlled) on the same soil specimen, either statically or dynamically, at up to 50 Hz.

With this system, soil specimens can be tested under isotropic (hydrostatic) or anisotropic conditions. The driving and measuring system are mounted on a "floating" frame, which allows large vertical specimen deformations throughout the complete test. Its unique configuration also allows for very large shear deformations of up to $\pm 20^\circ$.

Hollow Cylinder Tests

Dynamic Hollow Cylinder Test

By providing digital servo control of axial load, torque, confining pressure, internal pressure, and back pressure, the GCTS Dynamic Hollow Cylinder Testing System is capable of subjecting the soil specimen to any stress state and follow any stress/strain path.

The system is capable of simulating most field stress/strain path tests (static or dynamic), including plane strain, simple shear, and small shear strains. Other possible tests include: measurement of dynamic shear strength & deformation, liquefaction potential, shear modulus and damping ratio. The system is versatile in allowing the user to design custom procedures with operating frequencies of up to 30 Hz.



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Large-Scale Dynamic Hollow Cylinder System

The GCTS Large-Scale Dynamic Hollow Cylinder System is an advanced testing apparatus used for the development of constitutive models and the determination of strength parameters of isotropic and anisotropic soils. It is capable of performing stress path tests with controlled rotation of the principal stress directions. Granular soils with large particles can be tested using this system, as the variation of the radial strain over the wall thickness is negligible for large diameter specimens.

The system provides independent control of axial load, torque, confining pressure, inside pressure, and back pressure. All 5 axes can be controlled using feedback from any direct or calculated input, including more than 20 test parameters that are automatically computed by the GCTS software in real-time.

The Hollow Cylinder Apparatus is the only device capable of imposing three-dimensional stress states and control the rotation of the principal stress direction. The included software easily allows to realistically imitate in the laboratory the varied stress path conditions encountered in the field.



Frozen Soil Tests

Frozen Soil Triaxial System

The GCTS Frozen Soil Triaxial System was designed to study the formation of gas hydrates within soil samples and to characterize the geomechanical properties and response of hydrate-laden soils and rocks found mainly within ocean sediments on the continental margins and embedded in regions of permafrost.

This equipment is capable of performing isotropic and anisotropic consolidation, UU, CD, and CU triaxial tests with volume change or pore pressure measurements, stress/strain path tests, KO consolidation, cyclic loading including cyclic stress paths and other tests. This system is also capable of performing flexible-wall permeability tests. The system complies with ASTM D3999 and D5311 specifications.



Frozen Dynamic Hollow Cylinder/ True Triaxial System

The GCTS Frozen Dynamic Hollow Cylinder System is a highly advanced system that can perform hollow cylinder tests on soil samples at very low temperatures. This system has direct digital servo control of the axial load, torque, temperature, confining pressure, internal pressure, and back pressure. This allows to have accuracy not currently seen by systems on the market.

The system is capable of performing true triaxial tests on frozen specimens. This system can simulate most field stress/strain path tests (either static or dynamic) including plane strain, simple shear, and small shear strains. This system is also capable of performing other tests, including measurement of dynamic shear strength and deformation, liquefaction potential, shear modulus, and damping ratio, all at controlled temperature conditions.

Tests For Unsaturated Soils

Fredlund SWCC Device

The Fredlund SWCC Device is a simple and flexible unsaturated soil testing apparatus to measure the soil water characteristic curve while applying various stress states.

The device can be used to obtain the complete soil-water characteristic curve (SWCC) for a soil sample. The Fredlund SWCC device allows the operator to control matric suctions from near zero values up to 1,500 kPa, and is capable of applying one-dimensional loading, K_o , to a specimen with a diameter up to 75 mm. The device is economically priced and is a complete "turn-key" system.

The cell is constructed from stainless steel and includes the necessary plumbing and valves for periodic flushing and measuring of diffused air. Several different high air-entry-value (HAEV) ceramic stones can be easily interchanged. Dead weights can be used to apply normal stresses. An optional pneumatic load frame loader is available to apply larger stresses.

Through its convenient design, the apparatus allows the use of a single soil specimen to obtain the entire SWCC with any number of data points.



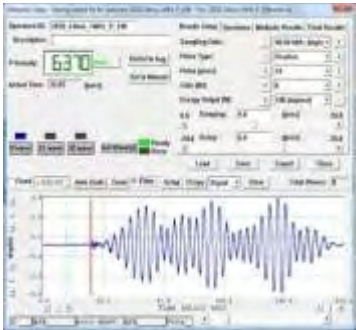


Soil Field & Laboratory

Unsaturated Soil Triaxial System

The GCTS Unsaturated Soil Triaxial System is an electro-pneumatic system with closed-loop digital servo control. Electro-hydraulic option is available for performing tests at higher frequencies or amplitudes. This system is developed for performing fully automated static and dynamic triaxial tests on saturated or unsaturated soil specimens. The system includes all the software modules and electrical valves required for automatic testing. The axial loader, cell pressure, pore air pressure, and pore water/back pressure are servo controlled using any system sensor as feedback.

The adoption of double cell walls essentially eliminates the compliance errors associated with single cell when measuring overall; specimen volume changes. This system performs unsaturated soil testing with a degree of automation not presently found in any other commercially available system. This system includes motorized ball valves to automatically recharge the pressure volume controllers; flush air bubbles diffused through the ceramic stone; and reverse the direction of the volume change device that enables infinite volume measurement range as required by the on-going test. This system also includes components such as a de-airing water reservoir and vacuum pump to perform saturated or unsaturated soil triaxial tests.



Ultrasonic P & S Wave Measurement System

The Ultrasonic Velocity Test System is a turnkey system and includes everything required to perform Ultrasonic Velocity measurements on laboratory specimens. The new hardware and software utilize the latest technology to allow for precise determination of wave velocities.

With the system the user also has the ability to control the pulse signal polarity and number of pulses to stack, this is important in helping to eliminate the influence of the Compressional wave signals on the relatively weaker Shear wave signals. The system uses a fast-acting pulser that provides excitation to the ultrasonic sensor and an ultra high speed Analog-to-digital converter for storing the resulting waveforms signals. The sampling rate can be selected from 40 MHz to a sampling rate as low as 156 Hz allowing the user to capture a wide range of ultrasonic signals

Erosion Function

Erosion Function Apparatus

The Erosion function Apparatus (EFA) was designed and built to prevent bridge failures by measuring the erodibility of soil. Used in conjunction with the SRICOS scour prediction method, the apparatus can provide more accurate erodibility measurements and scour predictions than previously obtainable. Applications for its use include: scour at bridges, piping of dams, beach erosion and surface erosion problems. In the case of scour at bridges, the EFA leads to improved accuracy on scour depth predictions, offering several advantages over previous test methods. These advantages include: minimum sample disturbance; measurement of erosion rate vs. shear stress; measurement of critical shear stress, and incorporation of the test results from the SRICOS scour prediction method. The erosion function apparatus uses standard 3.0" OD x 2.875" ID (76.2 x 73 mm) Shelby tubes. The SRICOS method improves the accuracy of pier scour predictions. The EFA is designed to be used in conjunction with the SRICOS method of scour prediction. The SRICOS scour prediction method and the erosion function apparatus were developed through research carried out by Jean-Louis Briaud, PHD, PE, and the scour research team at the Texas Transportation Institute of the Texas A&M University System. In comparison with the HEC-18 equation (a standard for calculating scour predictions), SRICOS generally leads to smaller calculated scour depths and compares more favorably to actual measured scour depths.



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Soil Processing, specimen preparation

Rapid Soil Processor

Compact all-metal soil processor allows customer to produce five-point moisture density relationship test results in about 5 hours. Processor also handles samples for rapid compaction of a family of curves test. Processes up to 30 lbs of lean-to-fat and tough clays at field moisture in fewer than 15 minutes. Automatically separates and retains up to .75" rocks from soil to prolong screen life. Processes soil to pass a No. 4 screen. Offset rotating drum is motor driven; pre-set adjustments aren't necessary. Automatic operation frees up technician's time. Order easy to-install replacement screen separately.



Soil Grinder

Prepares soil samples to designated particle size for accurate, repeatable test results. Grinds one pint sample in 15 seconds. Stainless steel construction. Includes a No. 10 perforated stainless plate.

Hobart Mixers - 5 Qt and 12 Qt

Mixer for mixing material samples. The mixers operate on the principle of planetary action where the beater reaches every part of a batch, rotating on its axis in opposite directions as it moves around the bowl. This mixer thoroughly blends, mixes and aerates all ingredients for a consistent, predictable finished batch. Selective agitator transmission has 3 speed settings. Direct gear drive and a heavy-duty motor ensure constant mixing speeds under load. A locking hand-lever provides precise raising and lowering of the mixing bowl.

Mechanical Compactor, Automatic

Meets ASTM D558, D559, D560, D698, D1557; AASHTO T99, T134, T135, T180. The mechanical compactor automatically compacts and rotates mold after each blow while keeping track of the number of hammer blows and shutting off once a preset number of blows is reached. The start/stop function of the compactor is independent of the counter. The unit can be used to perform standard or modified compaction tests using a 2.5kg hammer with 305mm height of drop or a 4.5kg hammer with 18" 457mm drop. Hammer lift compensates the height of the drop for soil thickness in the mold during compaction. Hammer weight is concentrated at the foot, allowing free fall of the hammer. Hammer changes are made from in front of the compactor.



Harvard Miniature Compaction Apparatus

Used for moisture density tests using small samples, its compaction action duplicates the kneading action of a sheep's foot roller. Mold is 33mm ID by 71mm H. Unit is designed so collar remover and specimen ejector are consolidated into one piece. Set includes specimen ejector, collar remover with spacer plate, moldholder, 129m3 volume mold and collar, compaction tamper with 9.07kg, 17kg and 18.2kg spring and operating instructions. Mold is machined from seamless tubing.



Test Equipment Atterberg Limits, Sand Equivalent, Hydrometer

We offer complete range of equipment and accessories for tests for liquid and plastic limits, Sand Equivalent, Hydrometer and other tests. Please contact us for information on any high quality soil laboratory equipment that you may require.